

Spin Dynamic Methods

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Contents

1	Notation and Conventions	1
2	Representative Material Parameter Sets	2
2.1	Property of Heavy Metal	2
2.2	Ferromagnet: CoFeB	3
2.3	Ferrimagnet: GdFeCo	3
2.4	Noncollinear Antiferromagnet: Mn ₃ Sn	5
3	Spin Hamiltonians	5
3.1	Atomistic Spin Hamiltonian	5
3.2	Macrospin Effective Fields	6
3.2.1	One-Sublattice Model	7
3.2.2	Two-Sublattice Collinear Model	8
4	Magnetization Dynamics	8
5	Thermal Stability	9
5.1	Thermal Stability Factor	9

1 Notation and Conventions

Unless a section explicitly states otherwise, vectors are bold, unit vectors carry a hat, time is measured in seconds, and magnetic fields $\mathbf{H}$ are expressed in A m⁻¹. The following symbols are used throughout the library.

$\mathbf{m}$ Normalized magnetization, with $\|\mathbf{m}\| = 1$.

$\mathbf{s}_i$ Unit spin direction at site i .

M_s Saturation magnetization of the ferromagnetic layer.

$\mathbf{H}_{\text{eff}}$ Effective magnetic field.

σ Spin-polarization unit vector.

α Dimensionless Gilbert damping parameter.

γ Positive gyromagnetic ratio, $\gamma = g\mu_B/\hbar$, where g is the Landé g -factor.

γ_0 Positive gyromagnetic factor for an $\mathbf{H}$ -field formulation, $\gamma_0 = \mu_0 |\gamma|$.

θ_{SH} Spin-Hall angle.

J_c Charge-current density.

t_{FM} Ferromagnetic-layer thickness.

β Signed torque ratio, $\beta = \tau_{\text{FLT}}/\tau_{\text{DLT}}$.

$\tau_{\text{DLT}}, \tau_{\text{FLT}}$ Damping-like and field-like torque amplitudes, respectively; both have field units.

e Positive elementary charge.

μ_0 Vacuum permeability.

μ_B Bohr magneton.

$\hbar$ Reduced Planck constant.

k_B Boltzmann constant.

T Absolute temperature.

Physical constants. The following values reproduce `constantfile.m`:

$$\begin{aligned}\gamma &= 1.760859644 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}, & e &= 1.602176565 \times 10^{-19} \text{ C}, \\ \mu_0 &= 1.25663706143592 \times 10^{-6} \text{ T m A}^{-1}, & \mu_B &= 9.274 \times 10^{-24} \text{ J T}^{-1}, \\ \hbar &= 6.58211951440 \times 10^{-16} \text{ eV s}, & k_B &= 1.38064852 \times 10^{-23} \text{ J K}^{-1}.\end{aligned}$$

Convention warning. Some references define the effective field as $\mathbf{B}_{\text{eff}}$ in tesla and use $|\gamma|$, whereas this library uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} and $\gamma_0 = \mu_0 |\gamma|$. Check this convention before copying an equation into simulation code.

2 Representative Material Parameter Sets

The values below reproduce particular numerical models from papers and repository implementations authored or coauthored by Zhifeng Zhu. They are model-specific parameter sets rather than universal material constants. In particular, quantities associated with a heavy-metal layer, spin polarization, device geometry, or numerical integration are identified separately from intrinsic magnetic parameters. Fixed-length unit vectors are used in all three source-model families summarized here.

2.1 Property of Heavy Metal

The spin-Hall angle belongs to the heavy-metal spin-current source and is not specific to a ferromagnetic, ferrimagnetic, or antiferromagnetic target layer. For the signed representative values in Table 1, we use $\mathbf{J}_s = \theta_{\text{SH}} \boldsymbol{\sigma} \times \mathbf{J}_c$ and choose Pt as the positive reference; Ta and W then have negative signs [5, 6, 7]. Reversing the current, layer-normal, or spin-polarization convention requires a corresponding sign change in the torque implementation.

These values are not universal elemental constants. In particular, Pai et al. find that W changes strongly with film phase and resistivity: a mixed α/β film gives $|\theta_{\text{SH}}| = 0.18 \pm 0.02$, whereas their predominantly α -W film gives only the upper bound $|\theta_{\text{SH}}| < 0.07$ [7]. The material-specific tables below omit θ_{SH} ; select a value from the heavy-metal source and sign convention used by the model.

The compact-model README compares reported resistivities of $190 \mu\Omega \text{ cm}$ for Ta and 170 – $260 \mu\Omega \text{ cm}$ for W, then uses the representative value $\rho_{\text{HM}} = 200 \mu\Omega \text{ cm}$ [5].

Property	Value	Evidence and qualification
Pt spin-Hall angle	+0.07	Representative literature value quoted by Liu et al. and Pai et al.; defines the positive reference sign
High-resistivity β -Ta spin-Hall angle	-0.12 to -0.15	Consistent magnitudes from ST-FMR, perpendicular switching, and three-terminal switching
High-resistivity β -W spin-Hall angle	-0.30 ± 0.02 (ST-FMR); -0.33 ± 0.06 (switching)	Negative sign verified relative to Pt; the ST-FMR value is a lower bound if spin diffusion suppresses the measured torque
Heavy-metal resistivity	$\rho_{\text{HM}} = 200 \times 10^{-8} \Omega \text{ m}$	Representative Ta/W value from the compact model; $\rho_{\text{HM}} = 200 \mu\Omega \text{ cm}$

Table 1: Representative properties of Pt, β -Ta, and β -W heavy metals [5, 6, 7].

2.2 Ferromagnet: CoFeB

The CoFeB free layer is represented by a one-sublattice macrospin LLGS model with perpendicular anisotropy. The anisotropy field and saturation magnetization below are the defaults of the SOT/STT compact model; the remaining magnetic and drive parameters come from the model of Zhang et al. [5, 4]. The compact-model parameter named $\mathbf{Hk}$ is a flux-density quantity in tesla. Its corresponding $\mathbf{H}$ field in A m^{-1} is obtained by dividing by μ_0 .

Quantity	Value	Role or convention
Uniaxial anisotropy field	$B_k = 1.5 \text{ T}$	Compact-model parameter $\mathbf{Hk}$; $H_k \approx 1.19 \times 10^6 \text{ A m}^{-1}$
Saturation magnetization	$M_s = 1000 \text{ emu cm}^{-3}$	$M_s = 1.00 \times 10^6 \text{ A m}^{-1}$; $\mu_0 M_s \approx 1.26 \text{ T}$
Gilbert damping	$\alpha = 0.02$	Magnetic model parameter

Table 2: Selected CoFeB macrospin parameters from the SOT/STT compact model and Zhang et al. [5, 4].

Parameter justification. Based on Nat. Mater. **9**, 721 (2010)., $H_{\text{an}} = 2(K_{\text{bulk}} + K_i/t_{\text{FL}})/M_s$ with $K_{\text{bulk}} = 2.245 \times 10^5 \text{ J m}^{-3}$, $K_i = 1.286 \times 10^{-3} \text{ J m}^{-2}$, $M_s = 1.58 \text{ T} = 1257 \text{ emu cm}^{-3}$, when we use $t_{\text{FL}} = 1.2 \text{ nm}$, it gives $H_{\text{an}} = 2.06 \text{ T}$. Here we approximate $M_s = 1000 \text{ emu cm}^{-3}$, $H_k = 1.5 \text{ T}$ [5].

2.3 Ferrimagnet: GdFeCo

Zhang et al. use a random two-dimensional atomistic $(\text{FeCo})_{1-x}\text{Gd}_x$ lattice driven by damping-like STT, whereas Zhu et al. use a one-dimensional atomistic GdFeCo model driven by SOT [13, 14]. Both are fixed-length LLG or LLGS models. The 2022 paper states that its parameters are otherwise the same as those in the 2020 paper, but replaces the single exchange coefficient with three bond-specific values. Table 3 therefore keeps the two published exchange parameterizations separately rather than selecting or averaging them.

The exchange values are deliberately not converted into a single common coefficient. The 2020 value belongs to a one-dimensional nearest-neighbor Hamiltonian, while the 2022 values distinguish the chemical identity of each bond in a random two-dimensional alloy. Their lattice topology and bond statistics must be retained with the numerical values when either set is used.

Quantity	Zhu et al. (2020)	Zhang et al. (2022)
Spatial model	1D atomistic lattice with lattice constant $d = 0.4$ nm; strip length and width are 400 and 40 nm	Random 2D square lattice; each interior site has four nearest neighbors; the main phase diagrams use 100 atoms
Exchange	$A_{\text{ex}} = +3$ meV = $+4.8065 \times 10^{-22}$ J	$A_{\text{Gd-Gd}} = -1.26 \times 10^{-21}$ J; $A_{\text{FeCo-FeCo}} = -2.83 \times 10^{-21}$ J; $A_{\text{FeCo-Gd}} = +1.09 \times 10^{-21}$ J
Perpendicular anisotropy	$K_{\text{TM}} = K_{\text{RE}} = 0.04$ meV	same as Zhu et al. (2020)
Hard-axis anisotropy	$\kappa_{\text{TM}} = \kappa_{\text{RE}} = 0.2$ μeV	Not included in the stated Hamiltonian
DMI	$D = 0.128$ meV	Not included in the stated Hamiltonian
Gilbert damping	$\alpha_{\text{TM}} = \alpha_{\text{RE}} = 0.015$	same as Zhu et al. (2020)
Landé factors	$g_{\text{TM}} = 2.2$; $g_{\text{RE}} = 2.0$	same as Zhu et al. (2020)
Sublattice magnetizations	$M_{s,\text{TM}} = 1149$ kA m $^{-1}$; $M_{s,\text{RE}} = 1012$ kA m $^{-1}$ at $T = 300$ K	same as Zhu et al. (2020)
Ferrimagnet thickness	$t_{\text{FiM}} = 0.4$ nm	same as Zhu et al. (2020)
Drive parameter	Spin-Hall angle $\theta_{\text{SH}} = 0.2$; field-like to damping-like ratio β is varied rather than fixed in the baseline list	STT efficiency η is introduced but no numerical value is stated
Compensation composition	Not stated as a composition in the parameter block	$x_{\text{MC}} = 0.23$; $x_{\text{AMC}} = 0.21$

Table 3: Atomistic GdFeCo parameter sets retained in their native model conventions [14, 13].

2.4 Noncollinear Antiferromagnet: Mn₃Sn

The three-sublattice macrospin switching model of Xu et al. and the later spatially resolved atomistic domain-wall model use the same magnetic moment, DMI, anisotropy, damping, and spin-Hall angle, but different exchange coefficients [8, 9]. The values must therefore remain attached to their model and interaction convention. Both Hamiltonians take positive A to favor antiparallel alignment, and the crystalline easy axes point toward the nearest Sn atoms. The atomistic domain-wall paper additionally states that its positive D favors clockwise order.

Quantity		Value	Model scope or convention
Mn moment		$m_s = 3\mu_B$	Per Mn site in both models
Nearest-neighbor exchange	ex-	$A = 17.53$ meV	Three-sublattice macrospin switching model
Nearest-neighbor exchange	ex-	$A = 9$ meV	Spatial atomistic domain-wall model
DMI coefficient		$D = 0.833$ meV	Same-layer interaction in both models
Crystalline anisotropy		$K = 0.196$ meV	Single-site Hamiltonian coefficient in both models
Gilbert damping		$\alpha = 0.003$	Same in both fixed-length models
Unit-cell volume		$V_{\text{cell}} = 0.3778$ nm ³	Macrospin model; $M_s = 6m_s/V_{\text{cell}} \approx 4.42 \times 10^5$ A m ⁻¹
Mn ₃ Sn thickness		$d = 30$ nm	Macrospin SOT model geometry
Time step		$\Delta t = 5$ fs	Fourth-order Runge–Kutta macrospin integration
Nanoribbon length		5.7 μm	Spatial atomistic domain-wall model geometry

Table 4: Mn₃Sn parameter sets used in the macrospin switching and atomistic domain-wall models [8, 9].

The Mn₃Sn entries are not by themselves a complete crystallographic simulation specification: a reproducible atomistic calculation must also retain the source model’s lattice, site mapping, bond list, DMI-vector orientations, boundary conditions, and pair-counting convention.

3 Spin Hamiltonians

3.1 Atomistic Spin Hamiltonian

Equation. Following the ASTRO exchange-sign convention, a classical spin Hamiltonian containing exchange, uniaxial anisotropy, Dzyaloshinskii–Moriya interaction, and Zeeman energy is

$$\begin{aligned}
\mathcal{H} = & + \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i K_i (\mathbf{s}_i \cdot \hat{\mathbf{e}}_i)^2 \\
& - \frac{1}{2} \sum_{i \neq j} \mathbf{D}_{ij} \cdot (\mathbf{s}_i \times \mathbf{s}_j) - \sum_i \boldsymbol{\mu}_i \cdot \mathbf{B}_{\text{ext}}.
\end{aligned} \tag{3.1}$$

Definitions. J_{ij} is the exchange-link energy in the ASTRO sign convention: $J_{ij} < 0$ favors parallel alignment, whereas $J_{ij} > 0$ favors antiparallel alignment. K_i is the uniaxial anisotropy energy, $\hat{\mathbf{e}}_i$ is the easy-axis direction, $\mathbf{D}_{ij}$ is the Dzyaloshinskii–Moriya vector, and $\boldsymbol{\mu}_i$ is the magnetic moment at site i .

Assumptions. Each $\mathbf{s}_i$ is a classical unit vector. The pair coefficients satisfy $J_{ji} = J_{ij}$ and $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$. The factors of 1/2 assume that the pair sums include both ordered pairs (i, j) and (j, i) . If each physical bond is instead included only once, for example in a unique-bond sum $\sum_{\langle i, j \rangle}$, both factors of 1/2 must be omitted. These are equivalent pair-counting conventions and must not be mixed.

Pair counting and effective field. Let $\mu_i = |\boldsymbol{\mu}_i|$ and define the effective flux density by

$$\mathbf{B}_{\text{eff},i} = -\frac{1}{\mu_i} \frac{\partial \mathcal{H}}{\partial \mathbf{s}_i}. \quad (3.2)$$

For the ordered-pair Hamiltonian above, the exchange and DMI contributions are

$$\mathbf{B}_{\text{ex},i} = -\frac{1}{\mu_i} \sum_{j \neq i} J_{ij} \mathbf{s}_j, \quad \mathbf{B}_{\text{DMI},i} = \frac{1}{\mu_i} \sum_{j \neq i} \mathbf{s}_j \times \mathbf{D}_{ij}. \quad (3.3)$$

The two ordered occurrences of each bond give identical derivatives, which cancel the Hamiltonian factors of 1/2. Consequently, no extra factor of 1/2 belongs in either effective-field expression.

Units. All terms in $\mathcal{H}$ have units of energy. Here the external flux density $\mathbf{B}_{\text{ext}}$ is measured in tesla, so no additional factor of μ_0 appears in the Zeeman term.

Source. For the extended classical spin Hamiltonian and Hamiltonian-derivative field convention, see Evans et al. [2].

Implementation notes. The ASTRO implementation evaluates the field, $\mathbf{B}_{\text{eff},i} = -(1/\mu_i) \partial \mathcal{H} / \partial \mathbf{s}_i$, rather than the total energy and includes both forward and backward nearest neighbors. The J_{ij} used above is therefore the coefficient stored in ASTRO's exchange-link arrays: its exchange field is $-(1/\mu_i) \sum_j J_{ij} \mathbf{s}_j$ [10]. No factor of 1/2 should appear in the exchange or DMI field formulas because differentiation produces two identical bond contributions, cancelling the Hamiltonian factor. ASTRO's DMI component expressions correspond to $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$ and the antisymmetry $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$ used above.

3.2 Macrospin Effective Fields

Field convention. For a fixed-length macrospin with normalized magnetization $\mathbf{m}$, let w denote the magnetic energy density in J m^{-3} . The effective magnetic field and the corresponding effective flux density are [16]

$$\mathbf{H}_{\text{eff}} = -\frac{1}{\mu_0 M_s} \frac{\partial w}{\partial \mathbf{m}}, \quad \mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}. \quad (3.4)$$

Thus, an LLG equation written with $\gamma_0 = \mu_0 |\gamma|$ uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} , whereas one written with $|\gamma|$ uses $\mathbf{B}_{\text{eff}}$ in tesla. The two conventions are physically equivalent but their field variables must not be mixed.

Unless stated otherwise, K_u and A_{12} are energy densities in J m^{-3} . The interfacial K_i in the combination $K_{\text{bulk}} + K_i/t_{\text{FM}}$ is instead in J m^{-2} , and every saturation magnetization is in A m^{-1} . The tensors $\mathbf{N}$ and $\mathbf{K}_{fp}$ are dimensionless.

3.2.1 One-Sublattice Model

Energy density and field decomposition. For uniaxial anisotropy with easy axis $\hat{\mathbf{u}}$, a dimensionless symmetric demagnetization tensor $\mathbf{N}$, an applied field $\mathbf{H}_{\text{ext}}$, and a dipolar field $\mathbf{H}_{\text{dip}}$ produced by another magnetic body, use the uniaxial-anisotropy convention from slides 13–14 of the EE237 Week 5 lecture notes [16] and take

$$w_1 = K_u \left[1 - (\mathbf{m} \cdot \hat{\mathbf{u}})^2 \right] + \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} - \mu_0 M_s \mathbf{m} \cdot (\mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{dip}}). \quad (3.5)$$

External field. The part of the final term in Eq. (3.5) that is proportional to $\mathbf{H}_{\text{ext}}$ is the Zeeman coupling to the applied field. Its derivative contributes $\mathbf{H}_{\text{ext}}$ directly to $\mathbf{H}_{\text{eff}}$ [16].

Uniaxial anisotropy field. Equation (3.4) gives [16]

$$\mathbf{H}_{\text{an}} = \frac{2K_u}{\mu_0 M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}, \quad \mathbf{B}_{\text{an}} = \frac{2K_u}{M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}. \quad (3.6)$$

For $\hat{\mathbf{u}} = \hat{\mathbf{z}}$, the anisotropy term reduces to $K_u(1 - m_z^2) = K_u \sin^2 \theta$, as on slide 13. Near $m_z = 1$, slide 14 therefore gives $H_{\text{an},z} = 2K_u/(\mu_0 M_s)$. For a thin film with bulk and interfacial uniaxial anisotropy, $K_u = K_{\text{bulk}} + K_i/t_{\text{FM}}$.

Demagnetization field. Slide 21 of the same notes writes the demagnetization self-energy as [16]

$$E_{\text{demag}} = -\frac{\mu_0}{2} \int_V \mathbf{M}(\mathbf{r}) \cdot \mathbf{H}_{\text{demag}}(\mathbf{r}) d^3r, \quad (3.7)$$

$$w_{\text{demag}} = -\frac{\mu_0}{2} \mathbf{M} \cdot \mathbf{H}_{\text{demag}} = \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} \quad (\text{uniform macrospin}). \quad (3.8)$$

The factor 1/2 avoids double counting the magnet's self-interaction. Equation (3.4) gives

$$\mathbf{H}_{\text{demag}} = -M_s \mathbf{N} \mathbf{m}, \quad \mathbf{B}_{\text{demag}} = -\mu_0 M_s \mathbf{N} \mathbf{m}. \quad (3.9)$$

For principal axes, $\mathbf{N} = \text{diag}(N_x, N_y, N_z)$ and $N_x + N_y + N_z = 1$.

Dipolar field. Let a pinned macrospin have volume V_p , saturation magnetization $M_{s,p}$, and direction $\mathbf{m}_p$, so its magnetic moment is $\boldsymbol{\mu}_p = V_p M_{s,p} \mathbf{m}_p$. In the point-dipole approximation, at displacement $\mathbf{r} = r \hat{\mathbf{r}}$ from the pinned layer to the free layer, slide 22 gives the dipole formula [16]. The slide denotes it by $\mathbf{H}_i$, but its factor μ_0 makes it a flux density in tesla; therefore, in the convention of this document,

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu}_p \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \boldsymbol{\mu}_p] = \mu_0 \mathbf{H}_{\text{dip}}, \quad (3.10)$$

Thermal field. Following Eq. (4) of Zhu et al. [11], write the sampled thermal field as

$$\mathbf{B}_{\text{th}} = \mu_0 \mathbf{H}_{\text{th}} = \mathcal{N}_1(0, u_B) \hat{\mathbf{x}} + \mathcal{N}_2(0, u_B) \hat{\mathbf{y}} + \mathcal{N}_3(0, u_B) \hat{\mathbf{z}}, \quad (3.11)$$

$$u_B = \sqrt{\frac{2\alpha k_B T}{V_{\text{FL}} M_s |\gamma| (1 + \alpha^2) \Delta t}}. \quad (3.12)$$

Here the $\mathcal{N}_i(0, u_B)$ are independent normal random variables with zero mean and standard deviation u_B , V_{FL} is the free-layer volume, and Δt is the duration over which a sampled field is held constant. With $|\gamma|$ in $\text{rad s}^{-1} \text{T}^{-1}$ and M_s in A m^{-1} , u_B is in tesla; the corresponding $\mathbf{H}$ -field standard deviation is $u_H = u_B/\mu_0$.

The source uses independent samples over $\Delta t = 5 \text{ ps}$ [11]. When adopting another integration interval or LLG convention, recompute the noise amplitude consistently rather than reusing that numerical value.

Total field. The total one-sublattice field, with the stochastic field added after taking the energy derivative, is [15]

$$\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{an}} + \mathbf{H}_{\text{demag}} + \mathbf{H}_{\text{dip}} + \mathbf{H}_{\text{th}}. \quad (3.13)$$

Spin-transfer and spin-orbit torques are nonconservative and are not part of the energy derivative in Eq. (3.4).

Implementation mapping. The current one-sublattice `ODE_LLГ_integral` implementation stores flux-density quantities in tesla [15]. Its `field_eta.m` evaluates the following expressions, where its input named `Hk` is B_k :

$$\begin{aligned} \mathbf{B}_{\text{an}} &= B_k m_y \hat{\mathbf{y}} \quad (\text{IMA}), & \mathbf{B}_{\text{an}} &= B_k m_z \hat{\mathbf{z}} \quad (\text{PMA}), \\ \mathbf{B}_{\text{demag}} &= -\mu_0 M_s \mathbf{N} \mathbf{m}, & \mathbf{B}_{\text{dip}} &= \mu_0 K_{12} M_s \mathbf{m}_{\text{PL}}, \end{aligned}$$

after converting the input M_s from emu cm^{-3} to A m^{-1} . It sums these with the applied and thermal fields.

3.2.2 Two-Sublattice Collinear Model

Inter-sublattice exchange field. Let $\mathbf{m}_1$ and $\mathbf{m}_2$ be separately normalized sublattice magnetizations with saturation magnetizations $M_{s,1}$ and $M_{s,2}$. The exchange-energy density convention used by the two-sublattice ferrimagnetic macrospin model of Yuan et al. [12] is

$$w_{\text{ex},12} = A_{12} \mathbf{m}_1 \cdot \mathbf{m}_2, \quad A_{12} > 0, \quad (3.14)$$

Applying $\mathbf{H}_{\text{ex},i} = -(\mu_0 M_{s,i})^{-1} \partial w_{\text{ex},12} / \partial \mathbf{m}_i$ yields, for $j \neq i$,

$$\mathbf{H}_{\text{ex},i} = -\frac{A_{12}}{\mu_0 M_{s,i}} \mathbf{m}_j, \quad \mathbf{B}_{\text{ex},i} = -\frac{A_{12}}{M_{s,i}} \mathbf{m}_j. \quad (3.15)$$

Only the exchange contribution unique to the two-sublattice model is stated here; the shared external, anisotropy, demagnetization, dipolar, and thermal field formulas are defined above and are not repeated. This two-sublattice exchange extension is publication-grounded and is not implemented by the `ODE_LLГ_integral` repository.

4 Magnetization Dynamics

LLGS form with field-like torque. This macrospin equation is adapted from Eqs. (1)–(4) of Kong et al., with the atom index removed [3]. It is written in the flux-density convention: $\mathbf{B}_{\text{eff}}$ and the torque amplitudes are in tesla and are paired with γ in $\text{rad s}^{-1} \text{T}^{-1}$.

$$\begin{aligned} \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (4.1)$$

Equivalent LL form with field-like torque. The algebraically equivalent form follows the same source convention [3].

$$\begin{aligned} (1 + \alpha^2) \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} - \alpha \gamma \mathbf{m} \times (\mathbf{m} \times \mathbf{B}_{\text{eff}}) \\ & + \alpha \gamma \tau_{\text{DLT}} \mathbf{m} \times \boldsymbol{\sigma} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \alpha \gamma \tau_{\text{FLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (4.2)$$

Torque amplitudes. For a ferromagnetic layer [4],

$$\tau_{\text{DLT}} = \frac{\hbar\theta_{\text{SH}}J_c}{2e t_{\text{FM}}M_s}, \quad \tau_{\text{FLT}} = \beta\tau_{\text{DLT}}.$$

Torque-sign convention. In the convention of Eq. (4.1), the sign product of the damping-like and field-like SOT is negative when $\beta > 0$ [1, 3]. Specifically, the scalar prefactors of their respective vector structures are $-\gamma\tau_{\text{DLT}}$ and $+\gamma\tau_{\text{FLT}}$, so their sign product is

$$(-\gamma\tau_{\text{DLT}})(+\gamma\tau_{\text{FLT}}) = -\gamma^2\beta\tau_{\text{DLT}}^2 < 0 \quad (\beta > 0).$$

Thus, a positive β means that the two SOT terms have oppositely signed scalar prefactors in the LLGS equation, even though the signed amplitudes τ_{DLT} and τ_{FLT} themselves have the same sign.

5 Thermal Stability

5.1 Thermal Stability Factor

Equation. For a single energy barrier E_b , the thermal stability factor is [17]

$$\Delta = \frac{K_u V}{k_B T} \tag{5.1}$$

Units. E_b and $k_B T$ must use the same energy unit; Δ is dimensionless.

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- [17] EE237 Week 10 lecture slides, MRAM Cells, Reliability, and Design Tradeoffs, slide 28.